

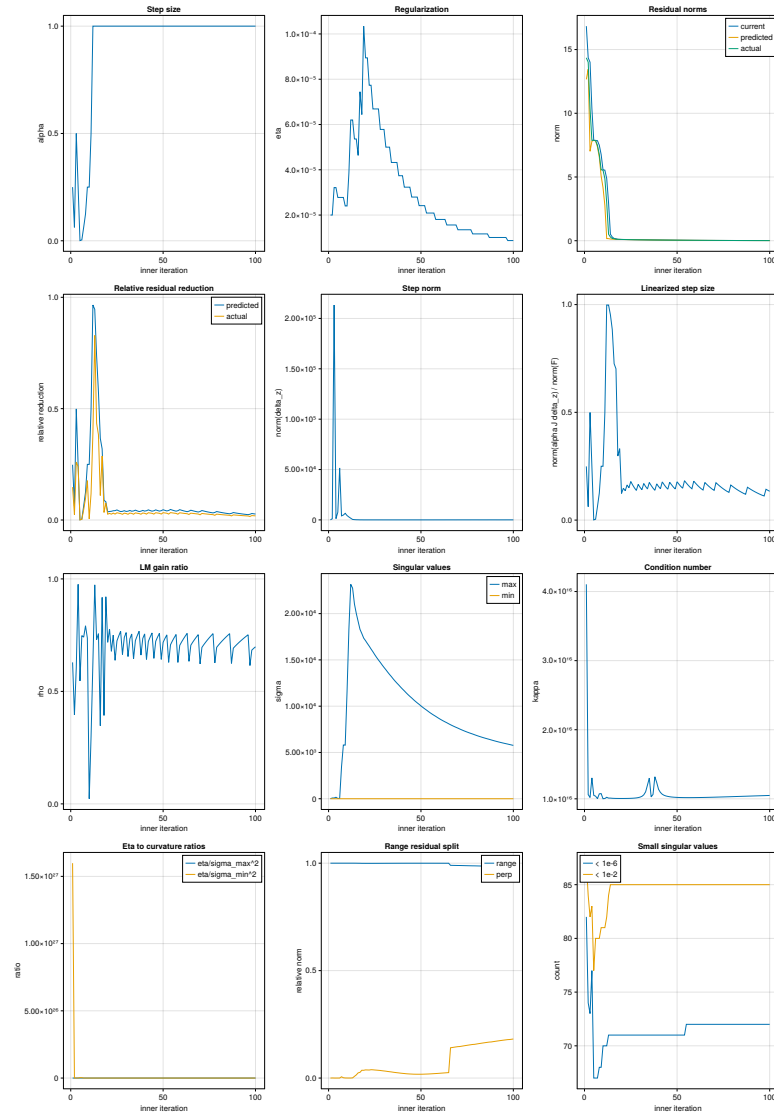
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000). Singular-value and range-space diagnostics use the raw ∇F SVD. If Marquardt column scaling is enabled for the step, the exact column space is unchanged, but numerical rank thresholding may differ.

Summary statistics

Metric	Minimum	Maximum	Mean	Median
alpha	9.765625e-04	1.000000e+00	9.125488e-01	1.000000e+00
eta	8.732233e-06	1.034460e-04	3.024953e-05	2.245856e-05
kkt_error	1.145437e-02	1.683624e+01	1.168830e+00	4.071875e-02
norm_delta_z	2.077556e+01	2.133356e+05	3.033369e+03	4.259645e+01
norm_alpha_J_delta_z / norm_F	9.757676e-04	9.980321e-01	2.029396e-01	1.522251e-01
predicted_residual_norm	1.114796e-02	1.345910e+01	8.664290e-01	3.890027e-02
predicted_relative_reduction	9.756499e-04	9.654218e-01	9.371153e-02	3.994992e-02
actual_residual_norm	1.124128e-02	1.435187e+01	1.000580e+00	3.940636e-02
actual_relative_reduction	5.330883e-04	8.288261e-01	5.824337e-02	2.852329e-02
lm_delta_pred	3.462742e-06	7.345429e+01	2.259809e+00	7.094108e-05
lm_delta_actual	2.290878e-06	4.442119e+01	1.417294e+00	4.871963e-05
lm_gain_ratio	2.281372e-02	9.759471e-01	7.000102e-01	7.245802e-01
count(Jsvd.S < 1e-6)	6.700000e+01	8.200000e+01	7.147000e+01	7.150000e+01
count(Jsvd.S < 1e-2)	7.700000e+01	8.700000e+01	8.457000e+01	8.500000e+01
maximum(Jsvd.S)	4.591747e+00	2.316496e+04	9.826615e+03	8.705655e+03
minimum(Jsvd.S)	1.118702e-16	2.266091e-12	9.487606e-13	8.552660e-13
condition_number	1.001169e+16	4.104531e+16	1.072832e+16	1.026622e+16
eta_over_sigma_max_sq	1.148854e-13	9.485802e-07	1.010517e-08	2.569122e-13
eta_over_sigma_min_sq	1.159119e+19	1.598090e+27	1.604910e+25	2.754900e+19
range_rank	3.960000e+02	4.110000e+02	4.006800e+02	4.000000e+02
norm(F_range) / norm(F)	9.833487e-01	1.000000e+00	9.951648e-01	9.995467e-01
norm(F_perp) / norm(F)	5.612347e-14	1.817286e-01	7.002052e-02	3.010248e-02
maximum(abs.(delta_z[1:144]))	9.224242e-04	4.420306e+00	1.906030e-01	4.789998e-03
median(abs.(delta_z[1:144]))	1.497935e-04	5.402795e-01	2.064208e-02	7.254584e-04
maximum(abs.(delta_z[145:end]))	1.220091e+01	2.066177e+05	2.817691e+03	4.204056e+01
median(abs.(delta_z[145:end]))	1.254924e-17	3.824504e-04	1.957083e-05	2.872102e-06

Metrics versus inner iteration



Per-iteration step diagnostics

Iter	α	η	$\ F\ $	$\ \delta z\ $	$\ \alpha J \delta z\ / \ F\ $	pred $\ F\ $	pred rel	actual $\ F\ $	actual rel	Δ_p	Δ_a	ρ
1	2.500000e-01	2.000000e-05	1.683624e+01	7.638804e+01	2.490550e-01	1.265938e+01	2.480873e-01	1.435187e+01	1.475608e-01	6.159951e+01	3.874139e+01	6.289236e-01
2	6.250000e-02	2.000000e-05	1.435187e+01	1.130932e+01	6.229707e-02	1.345910e+01	6.220591e-02	1.400582e+01	2.411177e-02	1.241442e+01	4.906577e+00	3.952320e-01
3	5.000000e-01	3.213061e-05	1.400582e+01	2.133356e+05	4.994664e+01	7.018156e+00	4.989116e-01	1.035957e+01	2.603383e-01	7.345429e+01	4.442119e+01	6.047461e-01
4	2.500000e-01	3.213061e-05	1.035957e+01	1.237804e+03	2.483257e-01	7.806687e+00	2.464275e-01	7.877808e+00	2.395624e-01	2.318817e+01	2.263043e+01	9.759471e-01
5	9.765625e-04	7.877808e+00	7.661221e+03	9.757676e-04	7.870122e+00	7.564996e-04	7.873608e+00	5.330883e-04	6.051915e-02	3.307456e-02	5.465140e-01	7.479729e-01
6	3.906250e-03	2.778221e-05	7.873608e+00	5.157579e+04	3.902681e-03	7.842883e+00	3.902216e-03	7.850638e+00	2.917312e-03	2.414408e-01	1.805912e-01	7.479729e-01
7	6.250000e-02	2.778221e-05	7.850638e+00	4.019490e+03	6.244642e-02	7.360453e+00	6.243896e-02	7.490377e+00	4.588938e-02	3.728130e+00	2.763384e+00	7.412253e-01
8	1.250000e-01	2.778221e-05	7.490377e+00	4.817072e+03	1.248979e-01	6.554962e-01	1.248823e-01	6.760623e+00	9.742560e-02	6.569112e+00	5.199866e+00	7.915629e-01
9	2.500000e-01	2.402229e-05	6.760623e+00	6.523398e+03	2.498275e-01	5.071838e+00	2.497973e-01	5.565860e+00	1.767237e-01	9.991241e+00	7.363609e+00	7.370065e-01
10	2.500000e-01	2.402229e-05	5.565860e+00	4.321632e+03	2.497055e-01	4.176267e+00	2.496637e-01	5.538047e+00	4.997232e-03	6.768797e+00	1.544215e-01	2.281372e-02
11	5.000000e-01	3.859255e-05	5.538047e+00	2.894766e+03	4.992694e-01	2.774339e+00	4.990401e-01	4.879773e+00	1.188639e-01	1.148650e+01	3.428889e+00	2.985147e-01
12	1.000000e+00	6.200012e-05	4.879773e+00	1.650351e+03	9.980321e-01	1.687337e-01	9.654218e-01	2.973293e+00	3.906903e-01	1.189186e+01	7.485855e+00	6.294943e-01
13	1.000000e+00	6.200012e-05	2.973293e+00	4.507949e+02	9.978700e-01	1.611548e-01	9.457992e-01	5.089501e-01	8.288261e-01	4.407290e+00	9.735596e-01	6.294943e-01
14	1.000000e+00	5.360932e-05	5.089501e-01	2.655164e+02	9.524789e-01	1.339513e-01	7.368086e-01	2.880710e-01	4.339897e-01	1.205436e-01	8.802265e-02	7.302141e-01
15	1.000000e+00	5.360932e-05	2.880710e-01	1.530282e+02	8.841992e-01	1.226161e-01	5.743544e-01	1.777073e-01	3.831127e-01	3.397508e-02	2.570250e-02	7.565102e-01
16	1.000000e+00	4.635408e-05	1.777073e-01	1.206101e+02	7.256210e-01	1.125308e-01	3.667634e-01	1.581537e-01	1.100328e-01	9.458357e-03	3.283651e-03	3.471692e-01
17	1.000000e+00	7.446926e-05	1.581537e-01	5.593070e+01	7.019459e-01	1.076961e-01	3.190415e-01	1.126885e-01	2.874749e-01	6.707070e-03	6.159505e-03	9.179791e-01
18	1.000000e+00	6.439094e-05	1.126885e-01	5.516186e+01	2.971362e-01	1.027069e-01	8.857645e-02	1.088753e-01	3.383778e-02	1.074989e-03	4.224255e-04	3.929578e-01
19	1.000000e+00	1.034460e-04	1.088753e-01	2.953549e+01	3.314597e-01	9.997443e-02	8.175326e-02	1.007090e-01	3.500630e-02	9.294770e-04	8.557682e-04	9.206987e-01
20	1.000000e+00	8.944612e-05	1.007090e-01	3.135806e+01	1.236516e-01	9.703049e-02	3.652617e-02	9.808115e-02	2.609354e-02	3.636938e-04	2.611958e-04	7.181749e-01
21	1.000000e+00	8.944612e-05	9.808115e-02	2.849161e+01	1.478388e-01	9.433318e-02	3.821294e-02	9.518248e-02	2.955386e-02	3.605815e-04	2.801044e-04	7.768129e-01
22	1.000000e+00	7.734091e-05	9.518248e-02	3.012942e-02	1.361988e-01	9.146462e-02	3.906025e-02	9.267707e-02	2.632215e-02	3.469631e-04	2.353324e-04	6.782634e-01
23	1.000000e+00	7.734091e-05	9.267707e-02	2.749755e+01	1.630221e-01	8.883443e-02	4.146266e-02	8.980976e-02	3.093872e-02	3.487415e-04	2.616232e-04	7.501922e-01
24	1.000000e+00	6.687395e-05	8.980976e-02	2.920577e+01	1.497808e-01	8.603683e-02	4.201027e-02	8.742104e-02	2.659753e-02	3.317286e-04	2.161772e-04	6.381034e-01
25	1.000000e+00	6.687395e-05	8.742104e-02	2.679165e+01	1.789929e-01	8.347164e-02	4.517676e-02	8.457881e-02	3.251198e-02	3.374617e-04	2.444317e-04	7.243242e-01
26	1.000000e+00	6.687395e-05	8.457881e-02	2.481681e+01	1.630371e-01	8.110133e-02	4.111520e-02	8.199573e-02	3.054053e-02	2.880743e-04	2.151378e-04	7.468137e-01
27	1.000000e+00	6.687395e-05	8.199573e-02	2.315241e+01	1.490723e-01	7.890056e-02	3.774793e-02	7.963083e-02	2.884170e-02	2.490006e-04	1.911150e-04	7.675281e-01
28	1.000000e+00	5.782355e-05	7.963083e-02	2.503567e+01	1.378481e-01	7.653780e-02	3.884213e-02	7.759050e-02	2.562238e-02	2.415172e-04	1.603918e-04	6.641009e-01
29	1.000000e+00	5.782355e-05	7.759050e-02	2.341083e+01	1.658898e-01	7.435154e-02	4.174434e-02	7.520644e-02	3.072624e-02	2.460674e-04	1.821389e-04	7.401991e-01
30	1.000000e+00	5.782355e-05	7.520644e-02	2.204656e+01	1.520903e-01	7.231587e-02	3.843515e-02	7.301548e-02	2.913256e-02	2.132118e-04	1.623738e-04	7.615611e-01
31	1.000000e+00	4.999798e-05	7.301548e-02	2.403517e+01	1.407755e-01	7.012481e-02	3.958983e-02	7.113596e-02	2.574141e-02	2.068857e-04	1.354679e-04	6.547957e-01
32	1.000000e+00	4.999798e-05	7.113596e-02	2.269405e+01	1.697085e-01	6.809252e-02	4.278341e-02	6.891644e-02	3.120112e-02	2.118667e-04	1.554247e-04	7.335966e-01
33	1.000000e+00	4.999798e-05	6.891644e-02	2.155416e+01	1.558885e-01	6.619646e-02	3.946773e-02	6.687255e-02	2.965753e-02	1.837519e-04	1.387689e-04	7.551974e-01
34	1.000000e+00	4.323149e-05	6.687255e-02	2.367583e+01	1.444889e-01	6.415217e-02	4.067998e-02	6.513234e-02	2.602277e-02	1.782181e-04	1.148580e-04	6.444801e-01
35	1.000000e+00	4.323149e-05	6.513234e-02	2.255955e+01	1.745279e-01	6.425305e-02	4.420668e-02	6.305471e-02	3.189857e-02	1.833894e-04	1.331625e-04	7.261191e-01
36	1.000000e+00	4.323149e-05	6.305471e-02	2.160178e+01	1.606711e-01	6.047908e-02	4.084748e-02	6.113899e-02	3.038181e-02	1.590884e-04	1.189599e-04	7.477600e-01
37	1.000000e+00	4.323149e-05	6.113899e-02	2.077556e+01	1.482712e-01	5.881663e-02	3.798490e-02	5.936517e-02	2.901295e-02	1.392900e-04	1.068765e-04	7.672950e-01
38	1.000000e+00	3.738074e-05	5.936517e-02	2.308982e+01	1.383054e-01	5.701698e-02	3.955500e-02	5.782288e-02	2.597974e-02	1.366437e-04	9.036907e-05	6.613484e-01
39	1.000000e+00	3.738074e-05	5.782288e-02	2.229548e+01	1.681003e-01	5.533869e-02	4.296199e-02	5.600647e-02	3.141327e-02	1.405572e-04	1.033801e-04	7.350255e-01
40	1.000000e+00	3.738074e-05	5.600647e-02	2.160646e+01	1.557129e-01	5.376603e-02	4.000325e-02	5.432229e-02	3.007121e-02	1.229694e-04	9.290687e-05	7.555284e-01
41	1.000000e+00	3.232181e-05	5.432229e-02	2.419708e+01	1.454035e-01	5.206407e-02	4.157071e-02	5.288324e-02	2.649097e-02	1.201217e-04	7.713708e-05	6.421577e-01
42	1.000000e+00	3.232181e-05	5.288324e-02	2.357177e+01	1.770662e-01	5.047762e-02	4.548936e-02	5.115822e-02	3.261949e-02	1.243237e-04	8.973701e-05	7.218013e-01
43	1.000000e+00	3.232181e-05	5.115822e-02	2.303049e+01	1.643380e-01	4.899186e-02	4.234628e-02	4.956012e-02	3.123820e-02	1.084806e-04	8.047852e-05	7.81705e-01
44	1.000000e+00	3.232181e-05	4.956012e-02	2.257848e+01	1.527871e-01	4.759673e-02	3.961646e-02	4.807541e-02	2.995777e-02	9.537871e-05	7.248028e-05	7.599209e-01
45	1.000000e+00	2.794753e-05	4.807541e-02	2.557944e+01	1.434948e-01	4.608422e-02	4.141815e-02	4.679719e-02	2.658784e-02	9.347507e-05	6.063410e-05	6.467977e-01
46	1.000000e+00	2.794753e-05	4.679719e-02	2.522106e+01	1.756629e-01	4.467206e-02	4.541146e-02	4.527090e-02	3.261494e-02	9.719198e-05	7.026121e-05	7.229116e-01
47	1.000000e+00	2.794753e-05	4.527090e-02	2.490706e+01	1.639057e-01	4.334792e-02	4.247735e-02	4.385295e-02	3.132147e-02	8.520648e-05	6.318664e-05	7.415708e-01
48	1.000000e+00	2.794753e-05	4.385295e-02	2.465751e+01	1.531413e-01	4.210337e-02	3.989659e-02	4.253274e-02	3.010547e-02	7.519388e-05	5.702380e-05	7.583569e-01
49	1.000000e+00	2.416524e-05	4.253274e-02	2.820062e+01	1.444849e-01	4.075316e-02	4.184033e-02	4.139866e-02	2.66358e-02	7.410711e-05	4.759226e-05	6.422091e-01
50	1.000000e+00	2.416524e-05	4.139866e-02	2.806620e+01	1.776906e-01	3.949182e-02	4.606059e-02	4.003884e-02	3.284710e-02	7.712289e-05	5.537042e-05	7.179505e-01
51	1.000000e+00	2.416524e-05	4.003884e-02	2.793423e+01	1.665394e-01	3.830873e-02	4.321086e-02	3.877389e-02	3.159302e-02	6.777505e-05	4.984700e-05	7.354771e-01
52	1.000000e+00	2.416524e-05	3.877389e-02	2.784097e+01	1.562521e-01	3.719663e-02	4.067853e-02	3.759512e-02	3.040124e-02	5.991281e-05	4.501091e-05	7.512736e-01
53	1.000000e+00	2.089483e-05	3.759512e-02	3.203534e+01	1.479596e-01	3.599019e-02	4.268977e-02	3.659385e-02	2.663280e-02	5.904951e-05	3.714135e-05	6.289865e-01
54	1.000000e+00	2.089483e-05	3.659385e-02	3.206211e+01	1.826710e-01	3.486341e-02	4.728784e-02	3.537869e-02	3.320674e-02	6.182640e-05	4.372918e-05	7.072897e-01
55	1.000000e+00	2.089483e-05	3.537869e-02	3.204598e+01	1.718435e-01	3.380692e-02	4.442707e-02	3.424807e-02	3.195761e-02	5.437198e-05	3.936064e-05	7.239141e-01
56	1.000000e+00	2.089483e-05	3.424807e-02	3.204236e+01	1.617889e-01	3.281431e-02	4.186414e-02	3.319458e-02	3.076068e-02	4.807588e-05	3.552521e-05	7.389404e-01
57	1.000000e+00	2.089483e-05	3.319458e-02	3.20463								

Iter	α	η	$\ F\ $	$\ \delta z\ $	$\ \alpha J \delta z\ / \ F\ $	pred $\ F\ $	pred rel	actual $\ F\ $	actual rel	Δ_p	Δ_a	ρ
84	1.000000e+00	1.167965e-05	1.585002e-02	4.767091e+01	1.329566e-01	1.537109e-02	3.021594e-02	1.549638e-02	2.231125e-02	7.476255e-06	5.542569e-06	7.413564e-01
85	1.000000e+00	1.167965e-05	1.549638e-02	4.698988e+01	1.280933e-01	1.504722e-02	2.898514e-02	1.516115e-02	2.163307e-02	6.859555e-06	5.138727e-06	7.491341e-01
86	1.000000e+00	1.167965e-05	1.516115e-02	4.631281e+01	1.235267e-01	1.473905e-02	2.784043e-02	1.484292e-02	2.098974e-02	6.310330e-06	4.774076e-06	7.565493e-01
87	1.000000e+00	1.009898e-05	1.484292e-02	5.259391e+01	1.198592e-01	1.440068e-02	2.979466e-02	1.456834e-02	1.849897e-02	6.466339e-06	4.037852e-06	6.244417e-01
88	1.000000e+00	1.009898e-05	1.456834e-02	5.178931e+01	1.523598e-01	1.408081e-02	3.346522e-02	1.423269e-02	2.303944e-02	6.983699e-06	4.833482e-06	6.921091e-01
89	1.000000e+00	1.009898e-05	1.423269e-02	5.092140e+01	1.467589e-01	1.377759e-02	3.197607e-02	1.391526e-02	2.230298e-02	6.373818e-06	4.467523e-06	7.009179e-01
90	1.000000e+00	1.009898e-05	1.391526e-02	5.006503e+01	1.413487e-01	1.348979e-02	3.057577e-02	1.361492e-02	2.158347e-02	5.830012e-06	4.134204e-06	7.091244e-01
91	1.000000e+00	1.009898e-05	1.361492e-02	4.922251e+01	1.362685e-01	1.321630e-02	2.927806e-02	1.333033e-02	2.090325e-02	5.347712e-06	3.834256e-06	7.169899e-01
92	1.000000e+00	1.009898e-05	1.333033e-02	4.839522e+01	1.314912e-01	1.295611e-02	2.807267e-02	1.306026e-02	2.025920e-02	4.918427e-06	3.563545e-06	7.245293e-01
93	1.000000e+00	1.009898e-05	1.306026e-02	4.758417e+01	1.269924e-01	1.270828e-02	2.695059e-02	1.280365e-02	1.964858e-02	4.535031e-06	3.318542e-06	7.317572e-01
94	1.000000e+00	1.009898e-05	1.280365e-02	4.679004e+01	1.227503e-01	1.247198e-02	2.590392e-02	1.255950e-02	1.906889e-02	4.191517e-06	3.096224e-06	7.386881e-01
95	1.000000e+00	1.009898e-05	1.255950e-02	4.601328e+01	1.187448e-01	1.224644e-02	2.492567e-02	1.232692e-02	1.851792e-02	3.882799e-06	2.893990e-06	7.453360e-01
96	1.000000e+00	1.009898e-05	1.232692e-02	4.525418e+01	1.149582e-01	1.203096e-02	2.400971e-02	1.210512e-02	1.799366e-02	3.604549e-06	2.709592e-06	7.517144e-01
97	1.000000e+00	8.732233e-06	1.210512e-02	5.130191e+01	1.118881e-01	1.179329e-02	2.575943e-02	1.191436e-02	1.575794e-02	3.726011e-06	2.290878e-06	6.148338e-01
98	1.000000e+00	8.732233e-06	1.191436e-02	5.040312e+01	1.430031e-01	1.156762e-02	2.910332e-02	1.167851e-02	1.979548e-02	4.071160e-06	2.782197e-06	6.833918e-01
99	1.000000e+00	8.732233e-06	1.167851e-02	4.947417e+01	1.382637e-01	1.135276e-02	2.789377e-02	1.145437e-02	1.919295e-02	3.751307e-06	2.592562e-06	6.911090e-01
100	1.000000e+00	8.732233e-06	1.145437e-02	4.857011e+01	1.336606e-01	1.114796e-02	2.675013e-02	1.124128e-02	1.860357e-02	3.462742e-06	2.418131e-06	6.983284e-01

Per-iteration SVD and range diagnostics

Iter	$\sigma_{\max}$	$\sigma_{\min}$	κ	$\eta/\sigma_{\max}^2$	$\eta/\sigma_{\min}^2$	rank	$\ F_r\ /\ F\ $	$\ F_{\perp}\ /\ F\ $	$N_{<10^{-6}}$	$N_{<10^{-2}}$
1	4.591747e+00	1.118702e-16	4.104531e+16	9.485802e-07	1.598090e+27	396	1.000000e+00	5.612347e-14	82	87
2	7.924901e+01	7.469383e-15	1.060985e+16	3.184508e-09	3.584764e+23	404	9.999999e-01	3.925855e-04	74	84
3	8.386632e+01	8.227705e-15	1.019316e+16	4.568186e-09	4.746370e+23	405	1.000000e+00	8.772789e-05	73	82
4	1.588648e+02	1.219395e-14	1.302816e+16	1.273103e-09	2.160876e+23	401	1.000000e+00	3.011639e-06	77	83
5	3.242323e+01	3.086407e-15	1.050812e+16	2.641257e-08	2.916489e+24	411	1.000000e+00	9.469944e-05	67	77
6	3.239576e+01	3.121338e-15	1.037880e+16	2.647223e-08	2.851577e+24	411	1.000000e+00	9.505830e-05	67	80
7	3.439848e+03	3.430715e-13	1.002662e+16	2.347947e-12	2.360464e+20	408	9.999827e-01	5.874718e-03	67	80
8	5.805681e+03	5.413866e-13	1.072372e+16	8.242528e-13	9.478763e+19	408	9.999990e-01	1.448444e-03	68	80
9	5.791877e+03	5.384887e-13	1.075580e+16	7.161035e-13	8.284406e+19	408	9.999999e-01	3.868940e-04	68	81
10	1.157532e+04	1.156180e-12	1.001169e+16	1.792869e-13	1.797063e+19	405	1.000000e+00	1.962821e-04	70	81
11	1.832818e+04	1.824685e-12	1.004457e+16	1.148854e-13	1.159119e+19	405	1.000000e+00	1.355668e-04	70	81
12	2.316496e+04	2.266091e-12	1.022243e+16	1.155392e-13	1.207363e+19	405	1.000000e+00	1.452810e-04	70	82
13	2.275083e+04	2.251119e-12	1.010645e+16	1.197838e-13	1.223476e+19	404	9.999997e-01	8.324270e-04	71	84
14	2.103563e+04	2.082011e-12	1.010351e+16	1.211517e-13	1.236728e+19	404	9.999684e-01	7.951153e-03	71	85
15	2.001567e+04	1.985867e-12	1.007906e+16	1.338135e-13	1.359377e+19	403	9.998933e-01	1.460998e-02	71	85
16	1.915116e+04	1.902430e-12	1.006669e+16	1.263856e-13	1.280769e+19	403	9.997078e-01	2.417438e-02	71	85
17	1.830015e+04	1.819521e-12	1.005767e+16	2.223657e-13	2.249379e+19	402	9.996480e-01	2.653071e-02	71	85
18	1.784499e+04	1.774901e-12	1.005408e+16	2.022051e-13	2.043979e+19	402	9.993404e-01	3.631541e-02	71	85
19	1.735555e+04	1.726578e-12	1.005199e+16	3.434292e-13	3.470096e+19	402	9.993479e-01	3.610673e-02	71	85
20	1.707782e+04	1.699011e-12	1.005163e+16	3.066878e-13	3.098627e+19	402	9.992785e-01	3.797947e-02	71	85
21	1.677120e+04	1.668406e-12	1.005223e+16	3.180044e-13	3.213349e+19	402	9.992915e-01	3.763625e-02	71	85
22	1.648343e+04	1.639498e-12	1.005395e+16	2.846520e-13	2.877315e+19	402	9.993032e-01	3.732522e-02	71	85
23	1.617038e+04	1.607796e-12	1.005748e+16	2.957799e-13	2.991901e+19	401	9.992467e-01	3.880882e-02	71	85
24	1.587786e+04	1.577844e-12	1.006301e+16	2.652607e-13	2.686141e+19	401	9.992632e-01	3.837964e-02	71	85
25	1.556089e+04	1.544907e-12	1.007238e+16	2.761775e-13	2.801899e+19	401	9.992972e-01	3.748533e-02	71	85
26	1.526585e+04	1.513568e-12	1.008600e+16	2.869559e-13	2.919130e+19	401	9.993225e-01	3.680374e-02	71	85
27	1.498984e+04	1.483367e-12	1.010528e+16	2.976206e-13	3.039204e+19	401	9.993504e-01	3.603879e-02	71	85
28	1.473090e+04	1.453843e-12	1.013239e+16	2.664688e-13	2.735710e+19	401	9.993798e-01	3.521476e-02	71	85
29	1.444981e+04	1.419760e-12	1.017764e+16	2.769366e-13	2.868632e+19	401	9.994215e-01	3.400857e-02	71	85
30	1.418764e+04	1.384559e-12	1.024704e+16	2.872663e-13	3.016349e+19	401	9.994547e-01	3.301858e-02	71	85
31	1.394194e+04	1.346218e-12	1.035637e+16	2.572210e-13	2.758809e+19	401	9.994875e-01	3.201294e-02	71	85
32	1.367551e+04	1.293407e-12	1.057324e+16	2.673411e-13	2.988700e+19	401	9.995300e-01	3.065688e-02	71	85
33	1.342723e+04	1.220268e-12	1.100350e+16	2.773191e-13	3.357699e+19	400	9.995634e-01	2.954808e-02	71	85
34	1.319475e+04	1.100884e-12	1.198559e+16	2.483122e-13	3.567116e+19	400	9.995948e-01	2.846473e-02	71	85
35	1.294288e+04	9.953901e-13	1.300282e+16	2.580704e-13	4.363285e+19	400	9.996333e-01	2.707874e-02	71	85
36	1.270835e+04	1.236543e-12	1.027733e+16	2.676835e-13	2.827365e+19	400	9.996628e-01	2.596855e-02	71	85
37	1.248891e+04	1.174216e-12	1.063595e+16	2.771733e-13	3.135482e+19	400	9.996894e-01	2.492224e-02	71	85
38	1.228298e+04	9.311361e-13	1.319139e+16	2.477652e-13	4.311432e+19	400	9.997133e-01	2.394573e-02	71	85
39	1.205946e+04	9.756833e-13	1.236002e+16	2.570348e-13	3.926722e+19	400	9.997407e-01	2.277140e-02	71	85
40	1.185088e+04	1.042281e-12	1.137014e+16	2.661625e-13	3.440952e+19	400	9.997609e-01	2.186526e-02	71	85
41	1.165533e+04	1.066789e-12	1.092562e+16	2.379285e-13	2.840130e+19	400	9.997782e-01	2.105959e-02	71	85
42	1.144328e+04	1.073436e-12	1.066042e+16	2.468283e-13	2.805069e+19	400	9.997977e-01	2.011609e-02	71	85
43	1.124555e+04	1.070163e-12	1.050826e+16	2.555842e-13	2.822251e+19	400	9.998106e-01	1.946360e-02	71	85
44	1.106033e+04	1.062147e-12	1.041319e+16	2.642160e-13	2.865013e+19	400	9.998208e-01	1.893287e-02	71	85
45	1.088634e+04	1.051859e-12	1.034962e+16	2.358193e-13	2.525970e+19	400	9.998285e-01	1.851688e-02	71	85
46	1.069732e+04	1.038613e-12	1.029962e+16	2.442269e-13	2.590814e+19	400	9.998371e-01	1.804830e-02	71	85
47	1.052071e+04	1.024965e-12	1.026445e+16	2.524953e-13	2.660266e+19	400	9.998407e-01	1.784719e-02	71	85
48	1.035497e+04	1.011305e-12	1.023922e+16	2.606429e-13	2.732622e+19	400	9.998423e-01	1.775813e-02	71	85
49	1.019901e+04	9.978766e-13	1.022071e+16	2.323139e-13	2.426819e+19	400	9.998422e-01	1.776577e-02	71	85
50	1.002931e+04	9.827562e-13	1.020528e+16	2.402422e-13	2.502071e+19	400	9.998424e-01	1.775439e-02	71	85
51	9.870471e+03	9.682512e-13	1.019412e+16	2.480364e-13	2.577597e+19	400	9.998385e-01	1.796902e-02	71	85
52	9.721169e+03	9.543475e-13	1.018619e+16	2.557138e-13	2.653249e+19	400	9.998334e-01	1.825449e-02	71	85
53	9.580472e+03	9.410442e-13	1.018068e+16	2.276487e-13	2.359494e+19	400	9.998271e-01	1.859471e-02	71	85
54	9.427161e+03	9.263530e-13	1.017664e+16	2.351132e-13	2.434927e+19	400	9.998214e-01	1.890134e-02	71	85
55	9.283443e+03	9.124331e-13	1.017438e+16	2.424492e-13	2.509787e+19	400	9.998118e-01	1.939907e-02	72	85
56	9.148160e+03	8.992091e-13	1.017356e+16	2.496729e-13	2.584149e+19	400	9.998015e-01	1.992440e-02	72	85
57	9.020502e+03	8.866342e-13	1.017387e+16	2.567896e-13	2.657969e+19	400	9.997905e-01	2.046694e-02	72	85
58	8.899768e+03	8.746635e-13	1.017508e+16	2.281021e-13	2.361591e+19	400	9.997791e-01	2.101984e-02	72	85
59	8.767758e+03	8.614950e-13	1.017738e+16	2.350225e-13	2.434339e+19	400	9.997686e-01	2.151109e-02	72	85
60	8.643551e+03	8.490371e-13	1.018042e+16	2.418256e-13	2.506302e+19	400	9.997542e-01	2.216911e-02	72	85
61	8.526242e+03	8.372138e-13	1.018407e+16	2.485257e-13	2.577590e+19	400	9.997395e-01	2.282445e-02	72	85
62	8.415201e+03	8.259746e-13	1.018821e+16	2.551277e-13	2.648215e+19	400	9.997245e-01	2.347339e-02	72	85
63	8.309878e+03	8.152740e-13	1.019274e+16	2.616359e-13	2.718188e+19	400	9.997092e-01	2.411449e-02	72	85
64	8.209789e+03	8.050709e-13	1.019760e+16	2.317770e-13	2.410272e+19	400	9.996937e-01	2.474696e-02	72	85
65	8.099841e+03	7.938282e-13	1.020352e+16	2.381121e-13	2.479027e+19	400	9.996800e-01	2.529808e-02	72	85
66	7.995879e+03	7.831638e-13	1.020972e+16	2.443441e-13	2.547001e+19	399	9.900239e-01	1.408992e-01	72	85
67	7.897250e+03	7.730183e-13	1.021612e+16	2.504855e-13	2.614296e+19	399	9.898267e-01	1.422783e-01	72	85
68	7.803499e+03	7.633508e-13	1.022269e+16	2.565403e-13	2.680934e+19	399	9.896325e-01	1.436231e-01	72	85
69	7.714228e+03	7.541247e-13	1.022938e+16	2.625121e-13	2.746933e+19	399	9.894410e-01	1.449361e-01	72	85
70	7.629083e+03	7.453071e-13	1.023616e+16	2.320799e-13	2.431709e+19	399	9.892522e-01	1.462195e-01	72	85
71	7.535217e+03	7.355698e-13	1.024405e+16	2.378979e-13	2.496516e+19	399	9.891804e-01	1.467039e-01	72	85
72	7.446126e+03	7.263086e-13	1.025201e+16	2.436247e-13	2.560588e+19	399	9.889587e-01	1.481911e-01	72	85
73	7.361312e+03	7.174771e-13	1.026000e+16	2.492710e-13	2.624013e+19	399	9.887402e-01	1.496426e-01	72	85
74	7.280433e+03	7.090426e-13	1.026798e+16	2.548401e-13	2.686813e+19	399	9.885255e-01	1.510544e-01	72	85
75	7.203186e+03	7.009759e-13	1.027594e+16	2.603352e-13	2.749008e+19	399	9.883144e-01	1.524290e-01	72	85
76	7.129299e+03	6.932506e-13	1.028387e+16	2.657593e-13	2.810616e+19	399	9.881069e-01	1.537685e-01	72	85
77	7.058529e+03	6.858431e-13	1.029175e+16	2.711151e-13	2.871657e+19	399	9.879027e-01	1.550748e-01	72	85
78	6.990656e+03	6.787318e-13	1.029958e+16	2.389979e-13	2.535324e+19	399	9.877018e-01	1.563495e-01	72	85
79	6.915427e+03	6.708446e-13	1.030854e+16	2.442260e-13	2.595291e+19	399	9.876178e-01	1.568795e-01	72	85
80	6.843613e+03	6.633068e-13	1.031742e+16	2.						

Iter	$\sigma_{\max}$	$\sigma_{\min}$	κ	$\eta/\sigma_{\max}^2$	$\eta/\sigma_{\min}^2$	rank	$\ F_{\gamma}\ /\ F\ $	$\ F_{\perp}\ /\ F\ $	$N_{<10^{-6}}$	$N_{<10^{-2}}$
84	6.585128e+03	6.361283e-13	1.035189e+16	2.693403e-13	2.886293e+19	399	9.864957e-01	1.637872e-01	72	85
85	6.526744e+03	6.299809e-13	1.036022e+16	2.741806e-13	2.942897e+19	399	9.862841e-01	1.650564e-01	72	85
86	6.470532e+03	6.240599e-13	1.036845e+16	2.789651e-13	2.999005e+19	399	9.860766e-01	1.662918e-01	72	85
87	6.416358e+03	6.183517e-13	1.037655e+16	2.453016e-13	2.641232e+19	399	9.858729e-01	1.674949e-01	72	85
88	6.356023e+03	6.119941e-13	1.038576e+16	2.499808e-13	2.696392e+19	399	9.857878e-01	1.679951e-01	72	85
89	6.298130e+03	6.058906e-13	1.039483e+16	2.545976e-13	2.750991e+19	399	9.855542e-01	1.693604e-01	72	85
90	6.242465e+03	6.000207e-13	1.040375e+16	2.591585e-13	2.805079e+19	399	9.853253e-01	1.706870e-01	72	85
91	6.188883e+03	5.943695e-13	1.041252e+16	2.636654e-13	2.858673e+19	399	9.851017e-01	1.719727e-01	72	85
92	6.137254e+03	5.889237e-13	1.042114e+16	2.681201e-13	2.911787e+19	399	9.848833e-01	1.732194e-01	72	85
93	6.087458e+03	5.836707e-13	1.042961e+16	2.725246e-13	2.964434e+19	399	9.846697e-01	1.744291e-01	72	85
94	6.039386e+03	5.785995e-13	1.043794e+16	2.768803e-13	3.016627e+19	399	9.844610e-01	1.756033e-01	72	85
95	5.992938e+03	5.736994e-13	1.044613e+16	2.811888e-13	3.068378e+19	399	9.842569e-01	1.767436e-01	72	85
96	5.948021e+03	5.689609e-13	1.045418e+16	2.854517e-13	3.119699e+19	399	9.840573e-01	1.778516e-01	72	85
97	5.904550e+03	5.643753e-13	1.046210e+16	2.504677e-13	2.741507e+19	399	9.838621e-01	1.789286e-01	72	85
98	5.855934e+03	5.592486e-13	1.047107e+16	2.546438e-13	2.792000e+19	399	9.837882e-01	1.793344e-01	72	85
99	5.809076e+03	5.543067e-13	1.047989e+16	2.587684e-13	2.842006e+19	399	9.835659e-01	1.805498e-01	72	85
100	5.763834e+03	5.495359e-13	1.048855e+16	2.628466e-13	2.891566e+19	399	9.833487e-01	1.817286e-01	72	85